

GLMs for Applied Research

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Welcome Session: Setup R/RStudio

Seminar schedule

Three days, 9 live sessions with hands-on practice in R

DAY 1

4:00 – 4:25 • Welcome session

Set Up: R & RStudio

4:30 – 5:00 • LECTURE

Session 1 • Why Go Beyond Linear Regression?

5:10 – 5:50 • LECTURE

Session 2 • Logistic Regression — Theory & Fitting

6:00 – 7:00 • HANDS-ON

Session 3 • Predicting a Binary Outcome

DAY 2

4:00 – 4:50 • LECTURE

Session 4 • Poisson Regression for Count Data

5:00 – 5:50 • HANDS-ON

Session 5 • Count Outcome Exercise

6:00 – 7:00 • LECTURE + LAB

Session 6 • Overdispersion & Negative Binomial Regression

DAY 3

4:00 – 4:50 • LECTURE + LAB

Session 7 • Multinomial & Ordinal Logistic Regression

5:00 – 5:50 • LECTURE

Session 8 • Introduction to Mixed-Effects GLMs

6:00 – 7:00 • HANDS-ON

Session 9 • Mixed-Effects models in R

R/RStudio set up

Complete this before the seminar/session 3 starts.

- **Install R**
Latest version, from cran.r-project.org
- **Install RStudio Desktop**
The free edition is all you need
- **Run 'Session_setup_Script.R' file**
Installs all R packages needed for the seminar
- **Bring basic R & tidyverse familiarity**
No prior GLM experience assumed — we build from here

How each session works

Sessions pair theory with hands-on practice

Lecture Slides

Theory
walkthroughs

Hands-on in R

Guided exercises on real
datasets

Live Q&A

Throughout each
session

Materials Shared

Slides/ Rmd files for
every session

Questions before or after the seminar? Reach out via the seminar forum.

Recommended literature

Optional reading to deepen each topic — not required to follow along

1. Agresti, A. (2013). *Categorical Data Analysis* (3rd ed.). Wiley.
2. Dobson, A. J., & Barnett, A. G. (2018). *An Introduction to Generalized Linear Models* (4th ed.). CRC Press/Chapman & Hall.
3. Faraway, J. J. (2016). *Extending the Linear Model with R: Generalized Linear, Mixed Effects and Nonparametric Regression Models* (2nd ed.). CRC Press.
4. Hilbe, J. M. (2011). *Negative Binomial Regression* (2nd ed.). Cambridge University Press. — *Overdispersion*
5. Beyond Multiple Linear Regression — Roback & Legler (free at bookdown.org/roback/bookdown-BeyondMLR)
6. An Introduction to Statistical Learning with Applications in R (2nd ed.) — James, Witten, Hastie & Tibshirani (free PDF at statlearning.com)
7. Modern Statistics with R — Måns Thulin (free, CC BY-NC-SA, at modernstatisticswithr.com)
8. Applied Statistics with R — David Dalpiaz (free at book.stat420.org)

What's next

Session 1: Why go beyond linear regression?